EXPERIMENT 12

Implementing Security Features and Governance / Administration Practices in Zoho Creator

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Aim

To implement and explore the security, access-control, data-sharing and governance features of a Zoho Creator application.

Introduction

Security and governance in low-code applications deal with controlling who can use an application, what each person is permitted to do, and how the data and resources of the application are administered. Because low-code platforms make it easy to build applications that store business data, controlling access to that data is an important part of application design.

Zoho Creator provides built-in mechanisms for this purpose. User permissions define the operations that a user may perform on the components of an application. Role-based access places users in roles that are arranged in a hierarchy, which influences the records that they can see. Data sharing rules extend the visibility of records from one role or user to another with a defined level of access. Administrative controls allow the application to be configured and managed. Together, they help ensure that users receive only the access needed for their responsibilities and that application data is protected from unauthorised use.

Application Used

The practical was carried out in Zoho Creator using the Event Registration application. The application is used for managing event registrations, and it served as the practical environment for exploring the security and governance features described in this record. The Zoho Creator account was on an expired trial ("Trial has expired" with an Upgrade option is displayed in the editor); no upgrade-only feature is claimed as configured in this record.

Security and Governance Architecture

Figure 1 is a conceptual supporting diagram of the security and governance structure of the Event Registration application. It is not a screenshot of the Zoho Creator interface.

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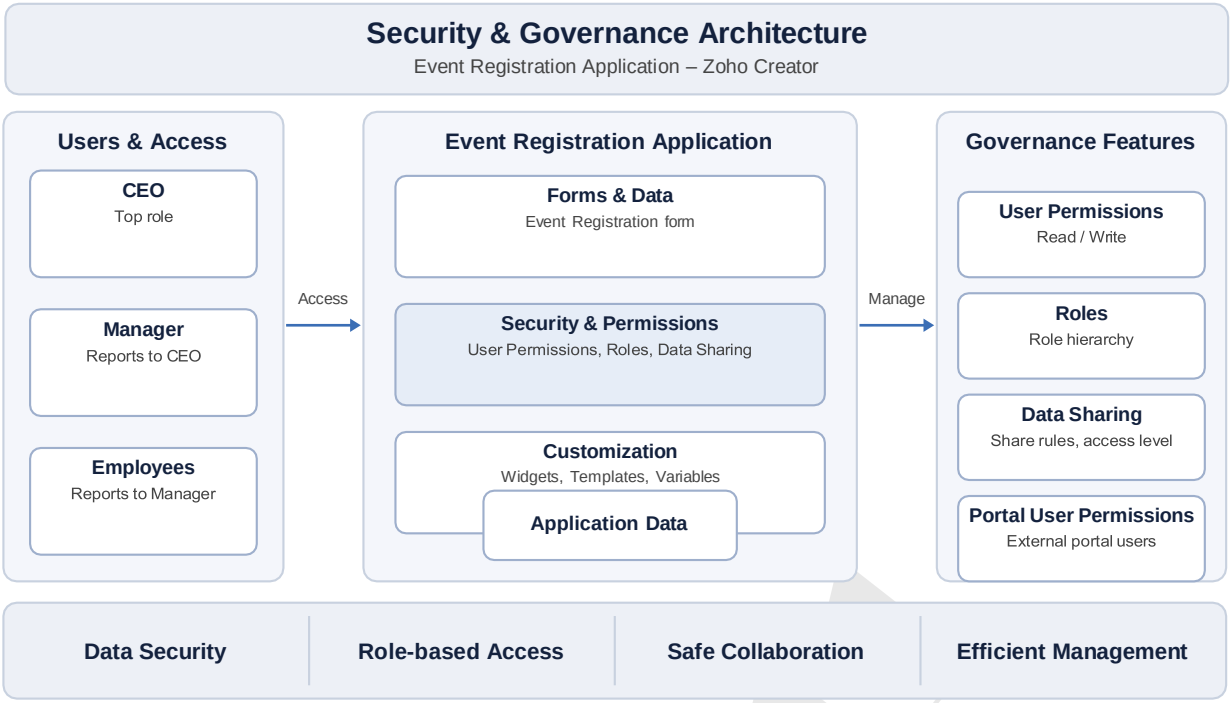


Figure 1: Security & Governance Architecture (Conceptual Diagram)

The diagram shows users (such as the CEO, manager and employees roles) accessing the application, which contains forms and data, security and permissions (user permissions, roles and data sharing) and customisation items. The governance features on the right (user permissions, roles, data sharing and portal user permissions) manage and control access to the application. The key outcomes are data security through controlled access, role-based access, safe collaboration through data sharing, and efficient management.

User Permissions

The User Permissions section is opened from Settings in the application editor. It contains the tabs Permissions, Roles and Data Sharing, and is used to add and manage the permissions and roles of users and to define the visibility of records for each user. The Permissions tab lists the permission sets available in the application and provides an Add Permission option. The application provided two predefined permissions:

Name	Description (as displayed)
Read	This profile will have read permission for all components
Write	This profile will have write permission for all components

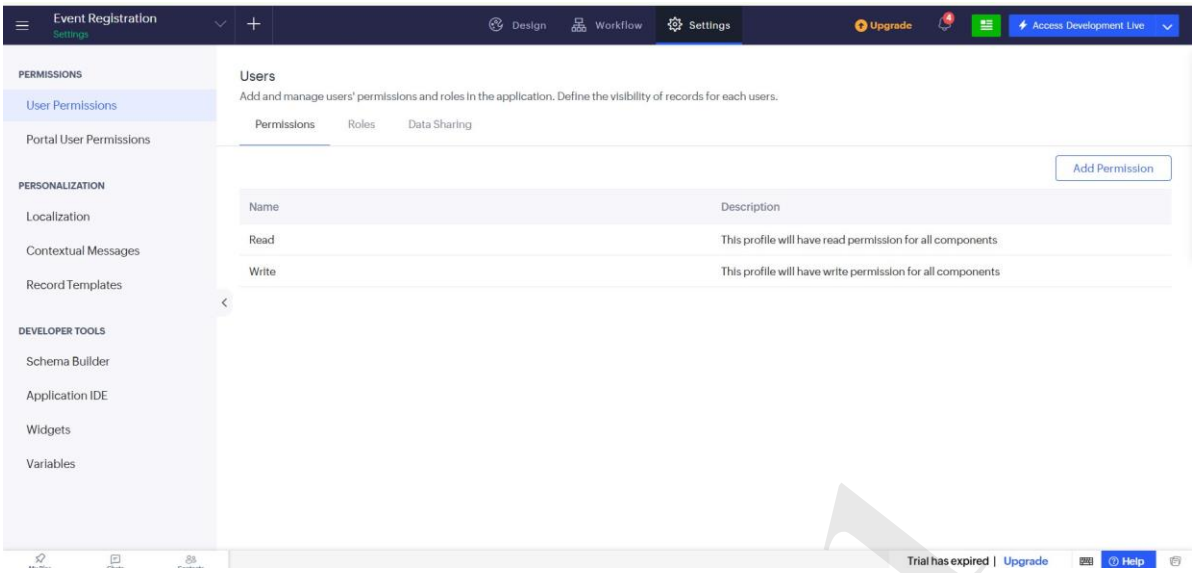


Figure 2: User Permissions in the Event Registration application

Read permission allows a user to view the components and records of the application without changing them, whereas Write permission allows the user to make changes, such as adding or modifying records.

Role-Based Access Control

The Roles tab shows the role hierarchy of the application. The hierarchy displayed is CEO → manager → employees, and the options Expand All, Collapse All, Manage Role Hierarchy and Add Role are available.

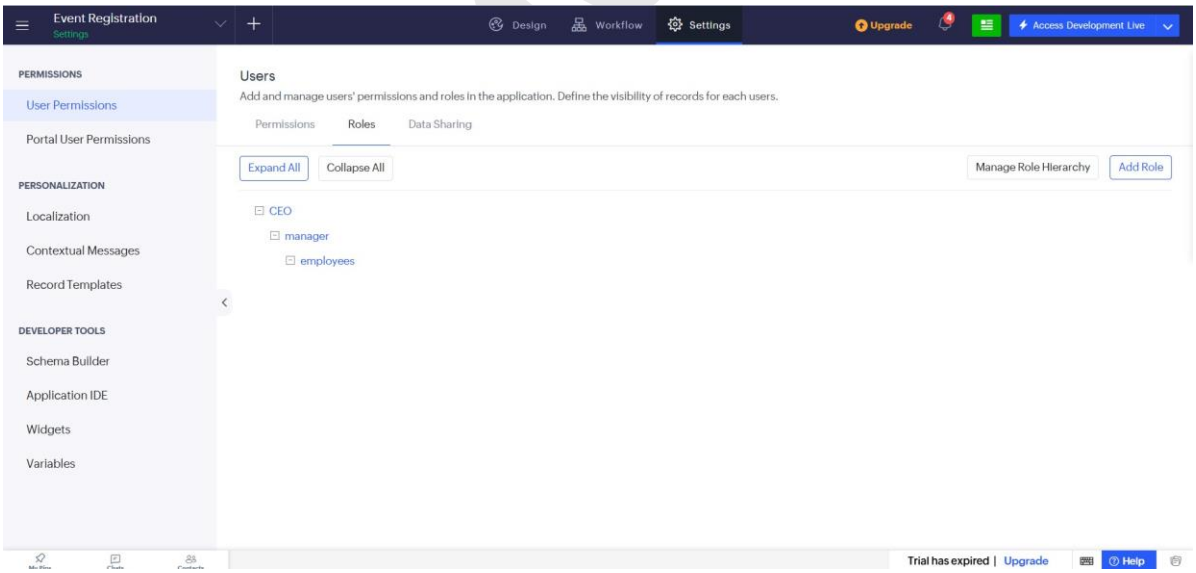


Figure 3: Roles and Role Hierarchy in the Event Registration application

Role	Position in hierarchy
CEO	Top-level role
manager	Reports to CEO
employees	Reports to manager

The Add Role option was used to prepare a custom role with the following details:

Field	Value
Role Name	technicians
Reports To	manager
Description	treat any bugs
Share Data with Peers	Selected (checkbox ticked)

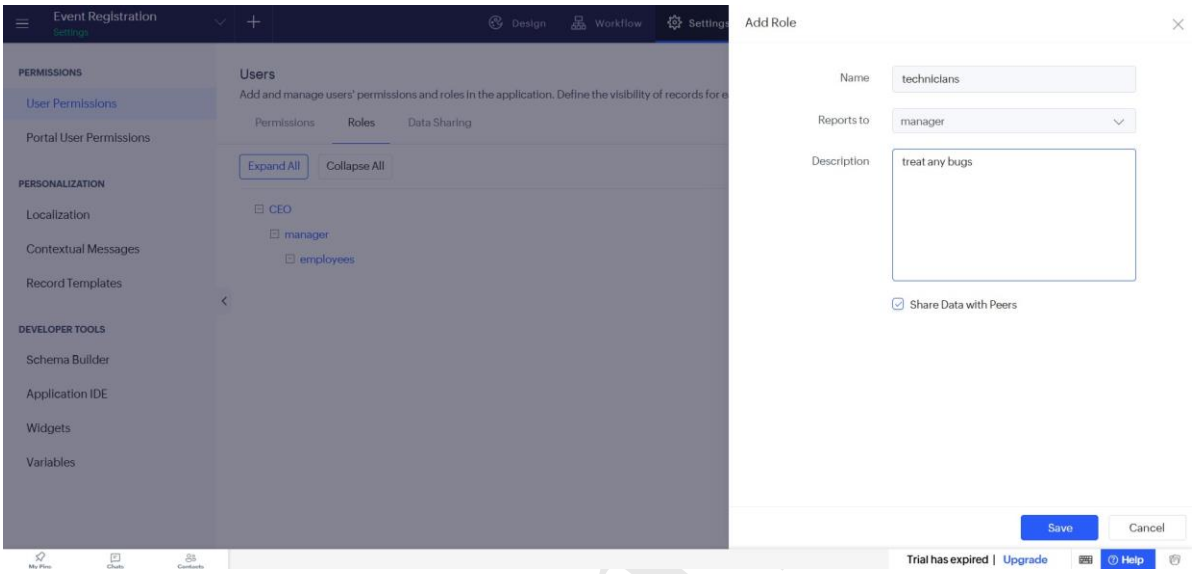


Figure 4: Add Role Panel – technicians Role Configuration

The technicians role was configured to report to the manager role, so that it is placed below manager in the hierarchy. With Share Data with Peers selected, users in this role can share their data with other users of the same role. The panel offers Save and Cancel options.

Data Sharing

Data sharing rules control the visibility of the records of a form. A rule specifies the form, the role or user from which records are shared, the role or user to which they are shared, and the level of access given. In this way, records can be made visible to additional roles or users beyond the default visibility of the role hierarchy. A data sharing rule was configured and appears in the Data Sharing list with the following details:

Parameter	Value
Rule Name	rule
Form	Event Registration
Shared From	RINUSWETHA (user)
Shared To	manager (role)
Access Level	Read/Write

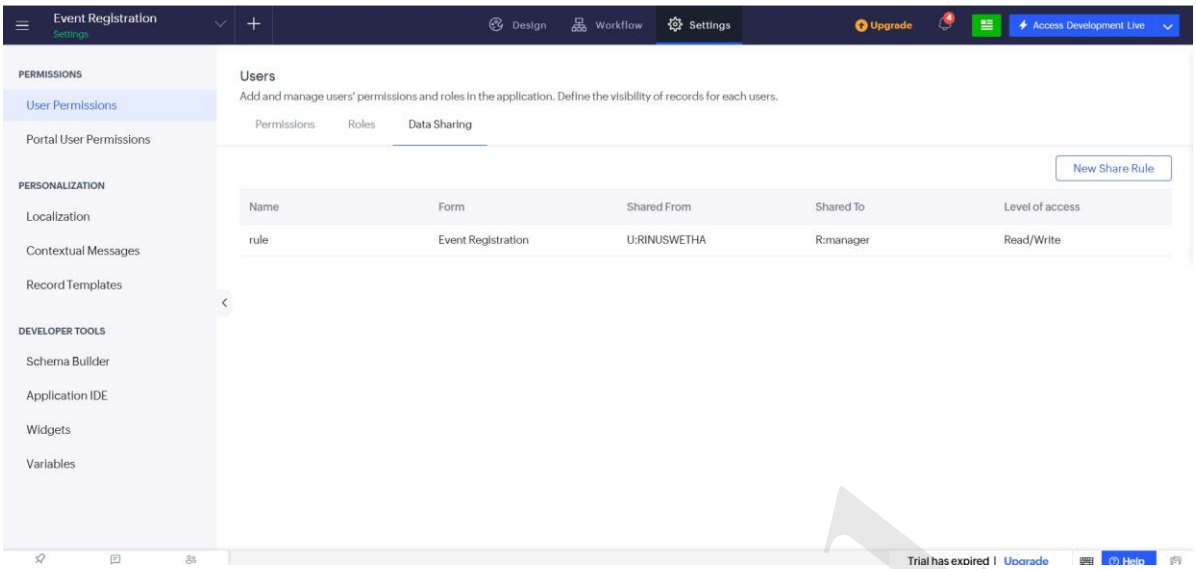
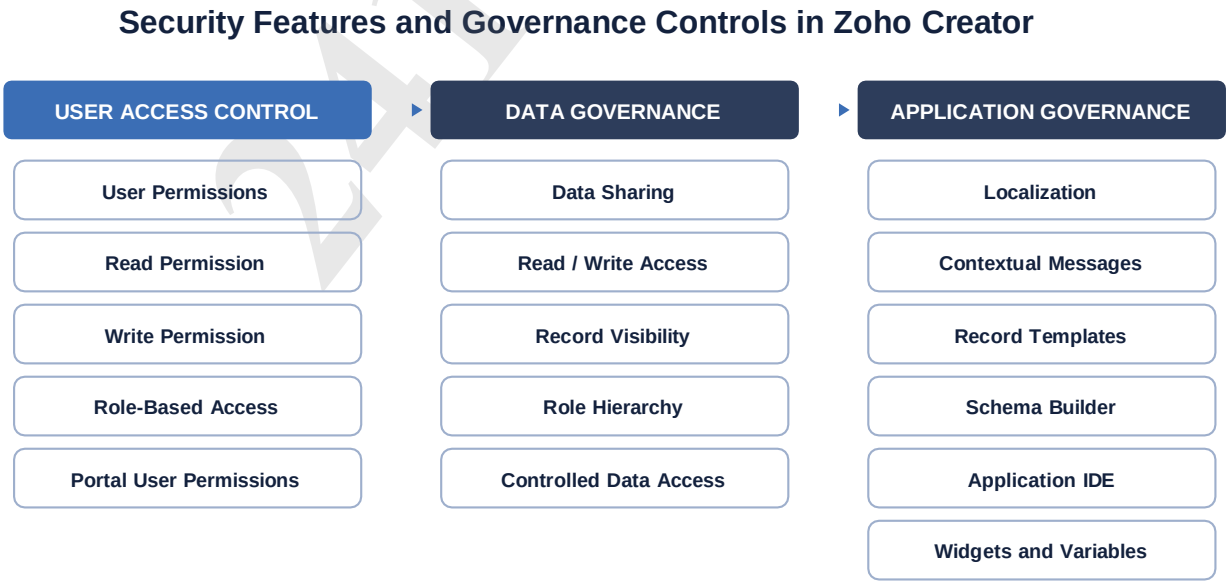


Figure 5: Data Sharing rule configured for the Event Registration form

The rule appears in the Data Sharing list with the columns Name, Form, Shared From, Shared To and Level of access. The prefixes U: and R: in the list indicate a user and a role respectively. With Read/Write access, users in the manager role can both view and modify the shared records of the Event Registration form that belong to the user RINUSWETHA. The level of access determines how much a recipient can do with the shared records, so that visibility is extended in a controlled manner.

Security Features and Governance Controls

Figure 6 is a conceptual academic diagram that groups the security and governance controls of Zoho Creator. It is not a screenshot, and it is not proof that every listed feature was configured.



Security and governance controls help manage who can access application data, what they can do, and how resources are administered.

Figure 6: Security Features and Governance Controls in Zoho Creator (Conceptual Diagram)

Control	Explanation
User access control	User permissions (Read and Write), role-based access and portal user permissions determine who can use the application and what they can do.
Control	**Explanation**
Data governance	Data sharing, read / write access, record visibility and role hierarchy govern how application data is accessed.
Application governance	Settings such as localization, contextual messages, record templates, schema builder, application IDE, widgets and variables are used to administer and customise the application.
Role hierarchy	Arranges roles in a reporting structure that influences which records a role can see.
Data sharing	Extends the visibility of records from one role or user to another with a chosen access level.
Record visibility	Decides which records are visible to which users, based on roles and sharing rules.
Administrative controls	Allow an administrator to manage permissions, roles and application settings.

In this practical, user permissions and roles were examined, a custom role configuration was prepared, and a data sharing rule was configured. The other items in the diagram, such as localization, schema builder and widgets, are listed in the Settings menu of the application, but they were not configured in this experiment.

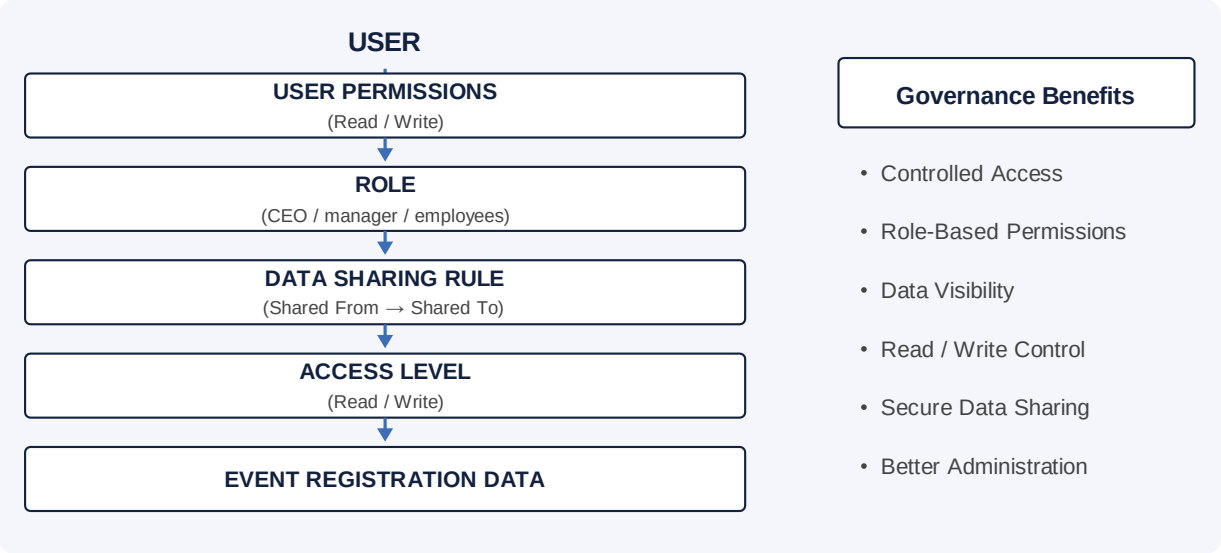
Portal User Permissions

Portal User Permissions is listed beside User Permissions in the Settings menu and is used to control the access of external portal users. It was seen in the Settings menu, but no portal permissions were configured in this experiment.

Role-Based Access and Data Sharing Flow

Figure 7 is a conceptual diagram of the security flow: User → User Permissions → Role → Data Sharing Rule → Access Level → Event Registration Data. It represents how access is controlled in principle and is not evidence that each step was configured.

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Role-based access and data-sharing rules help ensure users receive only the access required for their responsibilities.

Figure 7: Role-Based Access and Data Sharing Flow (Conceptual Diagram)

- A user is first given permissions (Read / Write).
- The user belongs to a role, such as CEO, manager or employees, which determines the position in the hierarchy.
- A data sharing rule specifies the records shared from one role or user to another.
- The access level (Read / Write) defines what the recipient can do, and this controls access to the Event Registration data.

Governance Observations

- Predefined Read and Write permissions were available in the application.
- The role hierarchy CEO → manager → employees could be viewed, and a custom role (technicians) configuration could be prepared in the Add Role panel.
- A data sharing rule was configured for the Event Registration form with Read/Write access.
- Portal User Permissions was listed in the Settings menu but was not configured.
- The account displayed "Trial has expired" with an Upgrade option; no upgrade-only feature is claimed as configured.

Conclusion

Zoho Creator provides built-in mechanisms for managing users, roles, permissions and data visibility. In this experiment, the predefined Read and Write permissions were examined, the role hierarchy of CEO, manager and employees was viewed, a custom role (technicians) configuration was prepared, and a data sharing rule was configured to share the records of the Event Registration form from the user RINUSWETHA to the manager role with Read/Write access.

Some advanced governance capabilities depend on the subscription plan, and the account displayed an expired trial with an Upgrade option. The experiment shows the access-control tools that Zoho Creator offers for the Event Registration application. Controlling who can access application data, and at what level, remains an essential part of building and managing business applications.

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